

Atharva Kirkole

East Lansing, Michigan | 5173560159 | kirkolea@msu.edu | [linkedin.com/in/atharva-kirkole](https://www.linkedin.com/in/atharva-kirkole) | github.com/AtharvaKirkole

EDUCATION

Michigan State University

Master of Science – Computer Science

Cumulative GPA: 3.9, **Teaching Assistant** – Object Oriented Principles, Parallel Computing (4.0)

East Lansing, MI

August 2025 – May 2026

Michigan State University

Bachelor of Science – Computer Science

Minor: Business, Honors College, GPA: 3.93, **Graduated with Honors**

All Semester Dean's List, Honors Societies: Tau Beta Pi and National Society of Scholars, Resident Assistant Leader

East Lansing, MI

August 2022 – April 2025

WORK EXPERIENCE

Michigan State University IT Services

Web Developer Intern

East Lansing, MI

June 2024 – Present

- Migrated **50+** university websites across CMS (**Sitecore**)
- Increased site accessibility score from **80% to 99.7%** by implementing **WCAG 2.1, ADA compliance, ARIA standards**
- Modernized university web services using **PHP, JavaScript, Docker**, and customized **Student data REST APIs**
- Led full migration from **Google Maps environment to ESRI ArcGIS platform** cutting IT expenses by ~20%

Amazon

Capstone Project Student

East Lansing, MI

January 2025 – May 2025

- Built **AI-powered semantic code search engine** for internal architecture assets using **AWS cloud services**
- Designed scalable backend using **Python, Node.js, AWS Lambda, DynamoDB, S3, and OpenSearch**
- Integrated **Amazon Bedrock and SageMaker** to generate vector embeddings for semantic similarity search
- Implemented vector search pipeline with embedding indexing and retrieval (RAG-style architecture).
- Developed **AI Employee recommendation system** using vector similarity and GitHub contribution analysis.
- Built responsive UI with **React and Cloudscape** and connected to RESTful APIs for real-time search.

Thompson Maize Lab

Machine Learning Researcher

East Lansing, MI

August 2023 – May 2024

- Designed scalable data architecture using **SQL + MongoDB** for large phenotypic and genomic datasets
- Built and evaluated **Random Forest and ML models (TensorFlow, Python, R)** for crop yield prediction
- Conducted statistical analysis on **Anthesis-Silking Interval (ASI)** and quantified yield impact under environmental stress
- Presented research at **North American Plant Phenotyping Network (NAPPN) Conference – Purdue University**
- **Published Abstract Link:** <https://www.authorea.com/users/530254/articles/675202>

PROJECTS

NanoGPT Attention Kernel Optimization (CUDA / OpenMP)

January 2026

- Implemented transformer attention in **C++** with contiguous tensor layouts and **4D accessors** to optimize memory locality.
- Applied **cache-aware tiling / blocking** for matrix to reduce DRAM traffic and improve L1/L2 reuse.
- Parallelized across batches/heads/rows with **OpenMP**, using thread-private buffers and CUDA kernel optimization.
- Vectorized hot loops using **SIMD intrinsics / ISPC (AVX2)** for linear algebra kernels on GPU.

Big Graph Processing in OpenMP (atomics/ OpenMP)

January 2026

- Implemented and optimized hybrid BFS for large, directed graphs, targeting high throughput on multi-core CPUs.
- Reduced synchronization overhead using **atomic operations / compare-and-swap** patterns to safely mark visited nodes under parallel frontier expansion.

AI Club –Navigator App

January 2023

- **Developed cross-platform Flutter + Flask indoor navigation app using CNN for image-based localization**
- Implemented A* pathfinding algorithm for real-time route optimization within large building layouts
- Awarded **People's Choice – MSU AI Club (Fall 2022)**

SKILLS

Pthreads, OpenCL, C++17, OpenMP, MPI, SIMD (AVX2), CPU cache/memory locality, profiling (perf/VTune/gprof), concurrency/atomics, Python, C++, JavaScript, SQL, R, PHP, HTML, CSS, React, Node.js, Flask, Django, .NET, RESTful APIs, Docker, AWS Lambda, S3, DynamoDB, AWS API Gateway, OpenSearch, Amazon Bedrock, MongoDB, PyTorch, TensorFlow, NLP, RAG, LLM's, CUDA, OpenMP, Google Maps API, Leaflet.js, ArcGIS, ESRI, ArcGIS Portal.

CERTIFICATIONS

- The Graduate ***Certificate in High Performance Computing*** at Michigan State University (Upon Graduation).
- CS50x HarvardX (Verified): <https://courses.edx.org/certificates/74771fe0522a47fb953b7ce7b8b2eb04>